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## Data Structure Assignment No:- 2

### Project Description:-

My Project is a Java-based FinTech trading System. It manages market assets such as stocks, cryptocurrencies and commodities. I used arrays to store assets information. For searching I implemented Linear & Binary Search. For sorting assets prices, I implemented Bubble sort, selection sort, insertion sort, Merge sort, and Quick sort. I used a singly linked list for the user watchlist because the assets can be inserted and deleted dynamically. I used recursion for displaying and calculating Portfolio value. I implemented a stack to manage transactions using LIFO and a Queue to manage trading orders using FIFO.

## Data Structure & Algorithms:-

### Arrays:-

Used to store market assets such as ID, type, Name, Price and Percentage change.

### Linear search:-

Used to search for an assets ID by checking assets one by one.

### Binary search:-

Used to search for an assets ID faster after arranging the IDs in sorted order.

### Bubble sort:-

Used to sort market assets according to their prices by comparing neighbouring elements.

### Selection sort:-

Used to sort assets prices by repeatedly selecting the smallest prices.

### Insertion sort:-

Used to assets prices by inserting each prices into its correct position.

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Merge sort:-

Used to sort assets prices by using divide & conquer.

Quick sort:-

Used to sort assets prices using a pivot and Partitioning.

Linked list:-

Used for watchlist where assets can be inserted and deleted, searched and displayed.

Recursion:-

Used to display portfolio assets and calculate total portfolio value. It is also used in quick and Merge sort.

Stack:-

Used to manage transaction. It follows LIFO.

Queue:-

Used to manage trading orders. It follows FIFO.

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Dry Run

Linear Search:

SearchID = 105

[Table]

Assets 105 was found at index 4.

Binary Search:-

SearchID = 107

[Table]

Asset 107 at index 6

Quick Sort:-

Initial:-

220.50, 250.75, 65000.00, 2650.50

Pivot: 2650.50

Final

2200.50 250.75, 2650.50, 65000.00.

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FINTECH TRADING SYSTEM
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1. Manage Market Assets
2. Search Asset
3. Sort Market Assets
4. Manage Watchlist
5. Calculate Portfolio Value
6. Manage Transactions
7. Manage Trading Orders
8. Exit

Enter your choice: 6